

Upper-Alias Counterloop Burden and Head-Transfer Precision

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Abstract

We determine the complete strict upper-alias burden of the source-locked finite-radius graded counterloop on the physical clock. Let $x = (\beta R)^2 > 1$, $y_k = (\beta_k R)^2$, and

$$\mathcal{C}_k^{\text{up}} = \sum_{2k < n < 4k} \frac{|s_{k,n}| R^n}{n}.$$

The exact counterloop ledger reduces this to $\sum_{m=k+1}^{2k-1} y_k^m / m$. The source-backed multiplier law $|M_k| = C_M \lambda^k \{1 + o(1)\}$ then gives

$$\mathcal{C}_k^{\text{up}} \sim \frac{x^{2k}}{2C_M^2 k(x-1)}, \quad x^{-k} \mathcal{C}_k^{\text{up}} \sim \frac{x^k}{2C_M^2 k(x-1)}.$$

Thus the normalized burden has k th root $x > 1$. Its terminal coordinate $n = 4k - 2$ contributes an asymptotic fraction $(x - 1)/x$.

We next state the exact price of actual-head transport. Conditional on the original, unnormalized same-clock hypothesis $D_{4k}(R) \rightarrow 0$, the actual upper-head budget has the same asymptotic, its odd upper budget tends to zero, every even upper coordinate has relative error $o(kx^{-k})$ uniformly, and the terminal coordinate has the stronger error $o(kx^{-2k})$. No theorem proving $D_{4k}(R) \rightarrow 0$ is supplied. A weaker normalized upper defect still transfers the aggregate normalized budget but does not force bandwise relative matching. A complete $(2k + 2)$ th-root shell gives a conjugation-closed finite normal information-class counterexample: its normalized defect is $\sim x/(C_M k) \rightarrow 0$, while the relative error at $n = 2k + 2$ equals one and the unnormalized defect diverges. This is not an actual noisy operator. No root or rank identification, direct/full-trace transfer, RH-288 activation, Gate A–E, Hilbert–Polya, zero-identification, or RH conclusion follows.

1 One physical clock and the exact upper band

Fix

$$r_H = \frac{17}{20}, \quad R = \frac{7}{5}, \quad \frac{28}{17} < \lambda < \frac{17}{10}. \quad (1)$$

Let $k = k_\sigma \rightarrow \infty$ be the physical first-alias clock used in RH-342. RH-17 proves the boundary multiplier law

$$|M_k| = C_M \lambda^k \{1 + o(1)\}, \quad C_M > 0, \quad (2)$$

and the finite counterloop radius is

$$\beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}. \quad (3)$$

Consequently

$$\beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right]. \quad (4)$$

Put

$$x = (\beta R)^2 = \frac{(R/r_H)^2}{\lambda}, \quad y_k = (\beta_k R)^2 = x \exp \left[-\frac{\log C_M}{k} + o(k^{-1}) \right]. \quad (5)$$

The certified upper bound for λ makes $x > 1$ [5, 7].

The counterloop is the finite multiset

$$\mathcal{Y}_k = \{\beta_k e^{ij\pi/k}, \beta_k e^{-ij\pi/k} : 1 \leq j \leq k-1\}. \quad (6)$$

RH-342 locks it in the same Hardy normalization as the actual modulus-complete head and proves the exact moment ledger [4]

$$s_{k,n} = \sum_{\nu \in \mathcal{Y}_k} \nu^n = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n). \quad (7)$$

For $2k < n < 4k$ there is no multiple of $2k$. Hence odd moments vanish and, for $k+1 \leq m \leq 2k-1$,

$$s_{k,2m} = -2\beta_k^{2m}, \quad \frac{|s_{k,2m}| R^{2m}}{2m} = \frac{y_k^m}{m}. \quad (8)$$

Define the strict upper-alias counterloop budget and its normalized version by

$$\mathcal{C}_k^{\text{up}} = \sum_{2k < n < 4k} \frac{|s_{k,n}| R^n}{n} = \sum_{m=k+1}^{2k-1} \frac{y_k^m}{m}, \quad \hat{\mathcal{C}}_k^{\text{up}} = x^{-k} \mathcal{C}_k^{\text{up}}. \quad (9)$$

RH-297 computed the isolated alias impulses at multiples of $2k$. Equation (9) instead sums every nonalias even coordinate in the strict upper band, so the result below is not a relabeling of that ledger [1].

2 The exponential counterloop burden

Theorem 2.1 (Strict upper-alias burden). *As $k \rightarrow \infty$,*

$$\mathcal{C}_k^{\text{up}} \sim \frac{x^{2k}}{2C_M^2 k(x-1)}, \quad (10)$$

$$\hat{\mathcal{C}}_k^{\text{up}} \sim \frac{x^k}{2C_M^2 k(x-1)}, \quad \lim_{k \rightarrow \infty} (\hat{\mathcal{C}}_k^{\text{up}})^{1/k} = x > 1. \quad (11)$$

For the terminal coordinate, put

$$\mathcal{T}_k = \frac{|s_{k,4k-2}| R^{4k-2}}{4k-2} = \frac{y_k^{2k-1}}{2k-1}. \quad (12)$$

Then

$$x^{-k} \mathcal{T}_k \sim \frac{x^{k-1}}{2C_M^2 k}, \quad \frac{\mathcal{T}_k}{\mathcal{C}_k^{\text{up}}} \longrightarrow \frac{x-1}{x}. \quad (13)$$

Proof. Set $m = 2k-1-r$ in (9). Then

$$\mathcal{C}_k^{\text{up}} = y_k^{2k-1} \sum_{r=0}^{k-2} \frac{y_k^{-r}}{2k-1-r}. \quad (14)$$

We first prove

$$k \sum_{r=0}^{k-2} \frac{y_k^{-r}}{2k-1-r} \longrightarrow \frac{x}{2(x-1)}. \quad (15)$$

For each fixed r , the summand after multiplication by k tends to $x^{-r}/2$. Choose $1 < \xi < x$. By $y_k \rightarrow x$, eventually $y_k \geq \xi$; also $k/(2k-1-r) \leq 1$ for $0 \leq r \leq k-2$. After extending the finite sum by zero, its terms are dominated by the summable sequence ξ^{-r} . Dominated convergence on the counting measure gives

$$\frac{1}{2} \sum_{r \geq 0} x^{-r} = \frac{x}{2(x-1)},$$

which is (15).

Equation (5) also gives

$$y_k^{2k-1} = x^{2k-1} C_M^{-2} \{1 + o(1)\}. \quad (16)$$

Substitute (15) and (16) into (14) to obtain (10). Division by x^k proves the first part of (11); taking k th roots proves the second. Finally, divide (16) by $2k-1$, normalize by x^k , and compare with (10). This gives both assertions in (13). $\square$

The theorem is unconditional for the deterministic graded counterloop. It does not yet say that an actual noisy-head budget is large or small: an actual head may match the shell by equally large complex moments.

3 What the original head-transport leaf would force

Let $\mathcal{H}_\sigma$ be the actual modulus-complete Hardy head of RH-342 and

$$h_{\sigma,n} = \sum_{\mu \in \mathcal{H}_\sigma} \mu^n, \quad d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}. \quad (17)$$

Suppress $\sigma = \sigma_k$ in the budgets and define

$$D_{4k}(R) = \sum_{2 \leq n < 4k} \frac{|d_{\sigma,k,n}| R^n}{n}, \quad (18)$$

$$\mathcal{H}_k^{\text{up}} = \sum_{2k < n < 4k} \frac{|h_{\sigma,n}| R^n}{n}, \quad (19)$$

$$\mathcal{H}_{k,\text{odd}}^{\text{up}} = \sum_{\substack{2k < n < 4k \\ n \text{ odd}}} \frac{|h_{\sigma,n}| R^n}{n}. \quad (20)$$

The quantity (18) is the original unnormalized, same-clock head/counterloop leaf in RH-288 and RH-340 [3, 6].

Theorem 3.1 (Conditional actual-head transfer precision). *Assume, on one physical clock, that*

$$D_{4k}(R) \longrightarrow 0. \quad (21)$$

Then

$$\mathcal{H}_k^{\text{up}} \sim C_k^{\text{up}} \sim \frac{x^{2k}}{2C_M^2 k(x-1)}, \quad (22)$$

$$\mathcal{H}_{k,\text{odd}}^{\text{up}} \longrightarrow 0. \quad (23)$$

Moreover, uniformly over $k+1 \leq m \leq 2k-1$,

$$\max_{k+1 \leq m \leq 2k-1} \frac{|h_{\sigma,2m} - s_{k,2m}|}{|s_{k,2m}|} = o(kx^{-k}), \quad (24)$$

and at the terminal coordinate

$$\frac{|h_{\sigma,4k-2} - s_{k,4k-2}|}{|s_{k,4k-2}|} = o(kx^{-2k}). \quad (25)$$

Proof. The reverse triangle inequality gives

$$|\mathcal{H}_k^{\text{up}} - \mathcal{C}_k^{\text{up}}| \leq D_{4k}(R) \longrightarrow 0. \quad (26)$$

Since theorem 2.1 gives $\mathcal{C}_k^{\text{up}} \rightarrow \infty$, (22) follows. Odd counterloop moments vanish in the strict upper band, so (20) is bounded by $D_{4k}(R)$, proving (23).

For an even upper order $n = 2m$, its weighted counterloop modulus is y_k^m/m . Since $y_k \rightarrow x > 1$, this sequence is increasing in $m \in [k+1, 2k-1]$ for all sufficiently large k . Thus

$$\max_{k+1 \leq m \leq 2k-1} \frac{|d_{\sigma,k,2m}|}{|s_{k,2m}|} \leq D_{4k}(R) \frac{k+1}{y_k^{k+1}}, \quad (27)$$

$$\frac{|d_{\sigma,k,4k-2}|}{|s_{k,4k-2}|} \leq D_{4k}(R) \frac{2k-1}{y_k^{2k-1}}. \quad (28)$$

From (5),

$$\frac{y_k^{k+1}}{x^{k+1}} \rightarrow C_M^{-1}, \quad \frac{y_k^{2k-1}}{x^{2k-1}} \rightarrow C_M^{-2}.$$

Now (21) means $D_{4k}(R) = o(1)$; inserting these two limits into (27) and (28) proves (24) and (25). Fixed powers of x are absorbed into the little- o notation. $\square$

Hypothesis (21) is not proved in this paper or inherited from RH-342. The theorem records a necessary precision obligation for any future proof of that original leaf; it is not actual head matching.

4 A normalized defect is strictly weaker

Define only the normalized strict-upper defect

$$\Delta_k^{\text{up}} = x^{-k} \sum_{2k < n < 4k} \frac{|d_{\sigma,k,n}| R^n}{n}. \quad (29)$$

If $\Delta_k^{\text{up}} \rightarrow 0$, then the same reverse triangle inequality and theorem 2.1 give

$$\frac{x^{-k} \mathcal{H}_k^{\text{up}}}{\hat{\mathcal{C}}_k^{\text{up}}} \longrightarrow 1. \quad (30)$$

At the terminal coordinate, (13) also forces

$$\frac{|h_{\sigma,4k-2} - s_{k,4k-2}|}{|s_{k,4k-2}|} = o(kx^{-k}). \quad (31)$$

This weaker hypothesis does not, however, control the bottom of the upper band, where a single normalized counterloop coordinate is only of order k^{-1} .

Proposition 4.1 (Complete-shell obstruction to bandwise matching). *Let*

$$N_k = 2k + 2, \quad a_k = \beta_k \left(\frac{2}{N_k} \right)^{1/N_k}, \quad (32)$$

and let $\mathcal{Z}_k$ be the complete N_k th-root shell of radius a_k . Write $z_{k,n} = \sum_{\zeta \in \mathcal{Z}_k} \zeta^n$ and form the finite multiset $\mathcal{H}_k^\# = \mathcal{Y}_k \cup \mathcal{Z}_k$, with moments $h_{k,n}^\# = s_{k,n} + z_{k,n}$. Then $\mathcal{H}_k^\#$ is finite and conjugation closed, and throughout $2k < n < 4k$ its defect is supported only at $n = N_k$. At that order,

$$|z_{k,N_k}| = |s_{k,N_k}| = 2\beta_k^{N_k}, \quad \frac{|h_{k,N_k}^\# - s_{k,N_k}|}{|s_{k,N_k}|} = 1. \quad (33)$$

Nevertheless,

$$x^{-k} \sum_{2k < n < 4k} \frac{|h_{k,n}^\sharp - s_{k,n}| R^n}{n} = \frac{x^{-k} y_k^{k+1}}{k+1} \sim \frac{x}{C_M k} \rightarrow 0, \quad (34)$$

$$\sum_{2 \leq n < 4k} \frac{|h_{k,n}^\sharp - s_{k,n}| R^n}{n} = \frac{y_k^{k+1}}{k+1} \sim \frac{x^{k+1}}{C_M k} \rightarrow \infty. \quad (35)$$

Thus normalized upper-defect convergence does not imply uniform bandwise relative matching or the original unnormalized $D_{4k}(R) \rightarrow 0$ leaf.

Proof. A complete N_k th-root shell has moment

$$z_{k,n} = N_k a_k^n \mathbf{1}_{N_k|n}.$$

Since $N_k \in (2k, 4k)$ and $2N_k = 4k + 4$, it is the only multiple of N_k in the strict upper band. Equation (32) gives $z_{k,N_k} = 2\beta_k^{N_k}$. Because N_k is even and is not divisible by $2k$ for $k \geq 2$, (7) gives $s_{k,N_k} = -2\beta_k^{N_k}$, proving (33). The weighted defect is

$$\frac{|z_{k,N_k}| R^{N_k}}{N_k} = a_k^{N_k} R^{N_k} = \frac{y_k^{k+1}}{k+1}.$$

Equations (34) and (35) now follow from $y_k^{k+1} = x^{k+1} C_M^{-1} \{1 + o(1)\}$. $\square$

Because $a_k \rightarrow \beta$ and $q < \beta < 1/r_H$, the added shell eventually lies inside the broad modulus window used for the Hardy head. This makes proposition 4.1 a legitimate finite normal information-class counterexample. It is not the spectrum of the actual noisy operator, does not satisfy an asserted actual rank law, and proves no actual nonmatching.

5 Claim boundary and reproducibility

RH-354 controls the actual direct coefficient

$$p_{\sigma,k,n} = \tau_{\sigma,n} - a_n = q_{\sigma,k,n} - d_{\sigma,k,n}. \quad (36)$$

Its normalized direct tail is not a theorem for q , d , or the full-trace E_{off} budget. Conversely, the counterloop asymptotic here does not close the direct coefficient. Transfer still requires the same-clock typed hypotheses of RH-340 and the remaining gluing leaves of RH-288 [2, 3, 6].

The executable artifact verifies the exact upper-band ledger at the rational fixture $\lambda = 5/3$, $C_M = 1$, and evaluates the synthetic exact multiplier law $|M_k| = C_M \lambda^k$ at archived high-precision diagnostics. It checks the two asymptotic ratios, the terminal share, and the complete-shell counterexample. These finite rows reproduce formulas only. The displayed decimal for C_M is not an interval certificate, and no row is an observation of the actual noisy head.

The unconditional theorem is solely about the source-locked graded counterloop. The actual-head conclusions are conditional on $D_{4k}(R) \rightarrow 0$, which remains open. No root matching, rank identification, physical determinant factorization, low-prefix closure, RH-241 envelope, RH-288 activation, or Gate A–E is proved. This work constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt-weighted trace or completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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